

The Complete Computables as the Minimal System for Financial Economics

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Abstract

Financial economics is traditionally formulated over the real numbers, the extended real numbers, or measure-theoretic probability spaces. This paper proposes an alternative computational foundation based on the notion of *Complete Computables*. The framework is motivated by the observation that ratio arithmetic naturally generates seven logically distinct valuation states: Null, Indeterminate, Zero, Positive, Negative, Positive Infinity, and Negative Infinity. We show that the resulting system is the minimal information-preserving representation capable of distinguishing all logically distinct security valuation outcomes. We further show that the extended real line may be constructed from Complete Computables while simultaneously appearing as a proper subset thereof.

The paper ends with “The End”

1 Introduction

Modern financial economics relies heavily upon ratio-based valuation. Examples include

$$\frac{S}{K}, \tag{1}$$

$$\ln\left(\frac{S}{K}\right), \tag{2}$$

$$\frac{\partial C}{\partial S}, \tag{3}$$

and

$$\frac{\partial^2 C}{\partial S^2}, \tag{4}$$

where S denotes the underlying asset price, K denotes strike, and C denotes option value.

Traditional formulations often conflate undefined, missing, infinite, and indeterminate states. The Complete Computables framework seeks to preserve all such distinctions.

2 Computational Foundations

2.1 Economic Currency Classes

We define four primitive economic states:

1. Worth-Null
2. Worth-Zero
3. Appreciating
4. Depreciating

These map naturally into computational price classes.

2.2 Complete Computables

Definition 1 (Complete Computables). *The Complete Computables are the seven-state system*

$$\mathcal{C} = \{\text{Null}, \text{Indeterminate}, 0, +, -, +\infty, -\infty\}. \quad (5)$$

Definition 2 (Symbolic Computables). *The Symbolic Computables are*

$$\mathcal{S} = \{\text{Null}, \text{Indeterminate}, 0, +, -\}. \quad (6)$$

Definition 3 (Naive Computables). *The Naive Computables are*

$$\mathcal{N} = \{\text{Null}, 0, +, -\}. \quad (7)$$

3 Ratio Algebra

Let

$$R(A, B) = \frac{A}{B}. \quad (8)$$

The Complete Computable ratio table is shown in Table 1.

Table 1: Complete Computable Ratio Algebra

A/B	Null	0	+	-
Null	Null	Null	Null	Null
0	Null	Indeterminate	0	0
+	Null	$+\infty$	+	-
-	Null	$-\infty$	-	+

4 Logical Structure

4.1 Information Ordering

The information hierarchy is

$$\text{Null} \prec \text{Indeterminate} \prec \{0, +, -\} \prec \{\pm\infty\}.$$

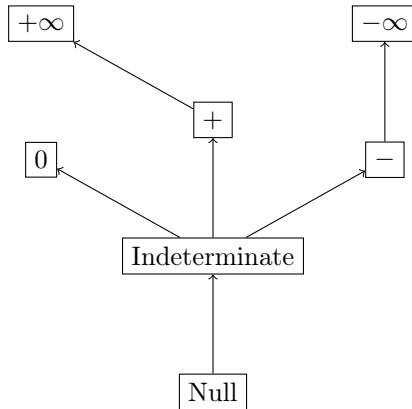


Figure 1: Information hierarchy of Complete Computables

5 Extended Reals

Definition 4 (Extended Reals).

$$\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}. \quad (9)$$

5.1 Construction from Complete Computables

The positive class expands as

$$+ \rightsquigarrow \mathbb{R}_{>0}, \quad (10)$$

while the negative class expands as

$$- \rightsquigarrow \mathbb{R}_{<0}. \quad (11)$$

Hence

$$\mathcal{C} \longrightarrow \overline{\mathbb{R}}. \quad (12)$$

5.2 Embedding

Conversely define

$$\pi : \overline{\mathbb{R}} \rightarrow \mathcal{C}$$

by

$$\pi(x) = \begin{cases} - & x < 0, \\ 0 & x = 0, \\ + & x > 0, \\ -\infty & x = -\infty, \\ +\infty & x = +\infty. \end{cases}$$

Therefore

$$\overline{\mathbb{R}} \subset \mathcal{C}. \quad (13)$$

6 Financial Economics

6.1 Security State Space

Every security valuation is represented by

$$(C, P),$$

where

- C denotes cost,
- P denotes price.

Definition 5. *A valuation system is complete if every logically distinct valuation state has a unique representation.*

Table 2: Financial Interpretation of Complete Computables

State	Financial Meaning
Null	No meaningful valuation exists
Indeterminate	Valuation undecidable
0	Worthless security
+	Positive asset value
-	Liability value
$+\infty$	Unbounded positive value
$-\infty$	Unbounded negative value

7 Option Pricing

Let

$$C(S, K, T, r, \sigma)$$

denote a generic option valuation function.
Boundary conditions include

$$\lim_{K \rightarrow 0^+} \frac{S}{K} = +\infty. \quad (14)$$

Likewise,

$$\frac{0}{0} = \text{Indeterminate}. \quad (15)$$

Thus option valuation naturally generates all Complete Computable states.

Theorem 1. *Neither Naive Computables nor Symbolic Computables can represent all option valuation states without information loss.*

Proof. Naive Computables identify

$$+\infty, -\infty, 0/0$$

with Null.

Symbolic Computables preserve

$$0/0$$

but collapse

$$+\infty \quad \text{and} \quad -\infty.$$

Therefore neither system preserves all valuation outcomes.

Complete Computables preserve every logically distinct ratio outcome. $\square$

8 Statistical Interpretation

Define a state random variable

$$X \in \mathcal{C}.$$

The entropy of a Complete Computable process is

$$H(X) = - \sum_i p_i \log p_i.$$

Because Complete Computables preserve more distinguishable states, they permit greater informational resolution than either Symbolic Computables or Naive Computables.

9 Machine Learning Interpretation

Suppose a learning system predicts valuation states.

Let

$$f_\theta : \mathcal{X} \rightarrow \mathcal{C}.$$

The target space becomes

$$\{\text{Null}, \text{Indeterminate}, 0, +, -, +\infty, -\infty\}.$$

The resulting classifier becomes a seven-class valuation learner.

10 Algorithms

10.1 Complete Computable Evaluation

Input: A, B

```
if A = Null or B = Null:
    return Null

if A = 0 and B = 0:
    return Indeterminate

if B = 0:
    if A > 0:
        return +Infinity
    else:
        return -Infinity

return sign(A/B)
```

11 Minimality Theorem

Theorem 2 (Minimality of Complete Computables). *The Complete Computables constitute the unique minimal information-preserving system capable of representing all logically distinct security valuation outcomes generated by ratio arithmetic.*

Proof. Ratio arithmetic generates the states

$$\{\text{Null}, \text{Indeterminate}, 0, +, -, +\infty, -\infty\}.$$

Removing any state forces at least two distinct valuation outcomes to be identified.

Hence every proper subset loses information.

Therefore Complete Computables are minimal and complete. $\square$

12 Conclusion

The Complete Computables framework extends traditional numerical representation by explicitly preserving nullity, indeterminacy, sign, and divergence. The resulting seven-state structure provides a minimal information-preserving foundation for ratio-based financial economics and option valuation.

References

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Glossary

Complete Computables

Seven-state computational valuation system.

Naive Computables

Four-state computational system.

Symbolic Computables

Five-state computational system.

Null

Absence of meaningful valuation.

Indeterminate

Undecidable valuation state.

Positive Infinity

Unbounded positive valuation.

Negative Infinity

Unbounded negative valuation.

Extended Reals

The set $\mathbb{R} \cup \{-\infty, +\infty\}$.

Security

Financial contract possessing a cost and a price.

Option

Derivative security whose value depends upon an underlying asset.

The End